# R5 Resultant forces A level only

Name: ________________

Class: ________________

Date: ________________

Time: **408 minutes**

Marks: **338 marks**

Comments:

## Q1.

The three forces $\mathbf{F}_1$ , $\mathbf{F}_2$ and $\mathbf{F}_3$ are acting on a particle.

$$\mathbf{F}_1 = (25\mathbf{i} + 12\mathbf{j})\ \text{N}$$
$$\mathbf{F}_2 = (-7\mathbf{i} + 5\mathbf{j})\ \text{N}$$
$$\mathbf{F}_3 = (15\mathbf{i} - 28\mathbf{j})\ \text{N}$$

The unit vectors **i** and **j** are horizontal and vertical respectively.
The resultant of these three forces is **F** newtons.

(a) (i) Find the magnitude of **F**, giving your answer to three significant figures.

**(2)**

(ii) Find the acute angle that **F** makes with the horizontal, giving your answer to the nearest 0.1°

**(2)**

(b) The fourth force, $\mathbf{F}_4$ , is applied to the particle so that the four forces are in equilibrium.
Find $\mathbf{F}_4$ , giving your answer in terms of **i** and **j**.

**(1)**

**(Total 5 marks)**

## Q2.

**In this question use $g$ = 9.8 m s$^{-2}$, giving your final answers to an appropriate degree of accuracy.**

The diagram shows a box, of mass 8.0 kg, being pulled by a string so that the box moves at a constant speed along a rough horizontal wooden board.

The string is at an angle of 40° to the horizontal.



The tension in the string is 50 newtons.




The coefficient of friction between the box and the board is $\mu$

Model the box as a particle.

(a) Show that $\mu$ = 0.83

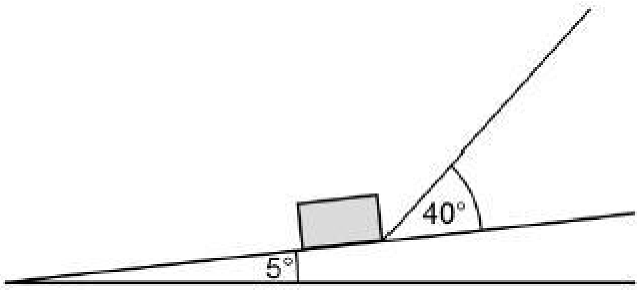
**(4)**

(b) One end of the board is lifted up so that the board is now inclined at an angle of 5°

to the horizontal.

The box is pulled up the inclined board.
The string remains at an angle of 40° to the board.

The tension in the string is increased so that the box accelerates up the board at $3 \text{ m s}^{-2}$




(i) Draw a diagram to show the forces acting on the box as it moves.

**(1)**

(ii) Find the tension in the string as the box accelerates up the slope at $3 \text{ m s}^{-2}$.

**(7)**
**(Total 12 marks)**

## Q3.

A box, of mass 3 kg, is placed on a rough slope inclined at an angle of 40° to the horizontal. It is released from rest and slides down the slope.

(a) Draw a diagram to show the forces acting on the box.

**(1)**

(b) Find the magnitude of the normal reaction force acting on the box.

**(2)**

(c) The coefficient of friction between the box and the slope is 0.2. Find the magnitude of the friction force acting on the box.

**(2)**

(d) Find the acceleration of the box.

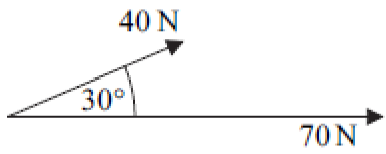
**(3)**

(e) State an assumption that you have made about the forces acting on the box.

**(1)**
**(Total 9 marks)**

## Q4.

Two forces, acting at a point, have magnitudes of 40 newtons and 70 newtons. The angle between the two forces is 30°, as shown in the diagram.




(a) Find the magnitude of the resultant of these two forces.

**(4)**

(b) Find the angle between the resultant force and the 70 newton force.

**(3)**
**(Total 7 marks)**

## Q5.

A block of mass 30 kg is dragged across a rough horizontal surface by a rope that is at an angle of 20° to the horizontal. The coefficient of friction between the block and the surface is 0.4.

(a) The tension in the rope is 150 newtons.

(i) Draw a diagram to show the forces acting on the block as it moves.

**(2)**

(ii) Show that the magnitude of the normal reaction force on the block is 243 newtons, correct to three significant figures.

**(3)**

(iii) Find the magnitude of the friction force acting on the block.

**(2)**

(iv) Find the acceleration of the block.

**(4)**

(b) When the block is moving, the tension is reduced so that the block moves at a constant speed, with the angle between the rope and the horizontal unchanged. Find the tension in the rope when the block is moving at this constant speed.

**(5)**

(c) If the block were made to move at a greater **constant** speed, again with the angle between the rope and the horizontal unchanged, how would the tension in this case compare to the tension found in part (b)?

**(1)**
**(Total 17 marks)**

## Q6.

A cyclist freewheels, with a constant acceleration, in a straight line down a slope. As the cyclist moves 50 metres, his speed increases from 4 m $s^{-1}$ to 10 m $s^{-1}$.

(a) (i) Find the acceleration of the cyclist.

**(3)**

(ii) Find the time that it takes the cyclist to travel this distance.

**(3)**

(b) The cyclist has a mass of 70 kg. Calculate the magnitude of the resultant force acting on the cyclist.

**(2)**

(c) The slope is inclined at an angle α to the horizontal.

(i) Find α if it is assumed that there is no resistance force acting on the cyclist.

**(3)**

(ii) Find α if it is assumed that there is a constant resistance force of magnitude 30 newtons acting on the cyclist.

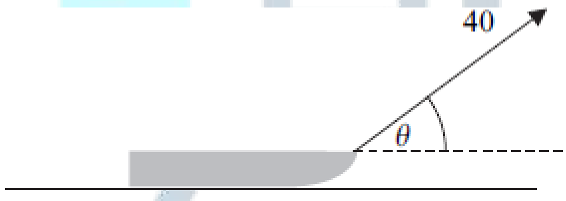
**(3)**

(d) Make a criticism of the assumption described in part (c)(ii).

**(1)**
**(Total 15 marks)**

## Q7.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope. The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is 40 newtons. The rope is kept at an angle of $\theta$ to the horizontal, as shown in the diagram.




Model the sledge as a particle.



(a) Draw a diagram to show all the forces acting on the sledge.

**(1)**

(b) Find the magnitude of the normal reaction force acting on the sledge, in terms of $\theta$.

**(3)**

(c) The acceleration of the sledge is

$$p + q \cos \theta + r \sin \theta$$

Find $p$, $q$ and $r$.

**(5)**
**(Total 9 marks)**

## Q8.

A stone, of mass 5 kg, is projected vertically downwards, in a viscous liquid, with an initial speed of 7 m s$^{-1}$.

At time $t$ seconds after it is projected, the stone has speed $v$ m s$^{-1}$ and it experiences a resistance force of magnitude $9.8v$ newtons.

(a) When $t \geq 0$, show that

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -1.96(v - 5)$$

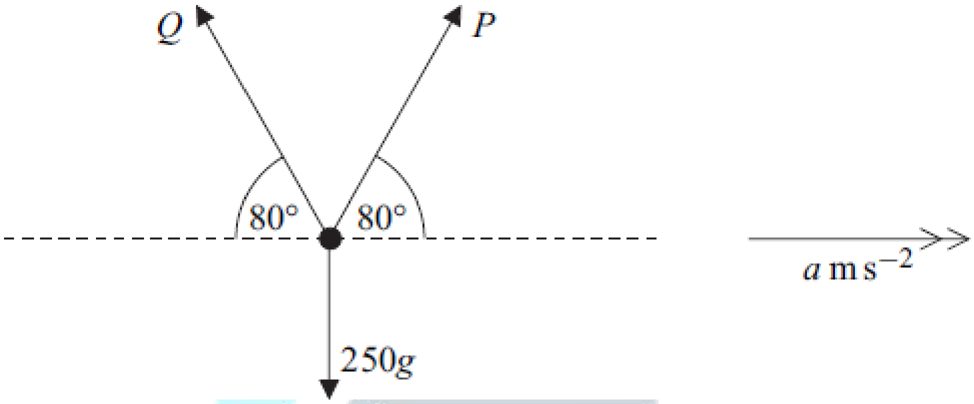
**(2)**

(b) Find $v$ in terms of $t$.

**(5)**
**(Total 7 marks)**

## Q9.

Three forces act in a vertical plane on an object of mass 250 kg, as shown in the diagram.




The two forces $P$ newtons and $Q$ newtons each act at 80° to the horizontal. The object accelerates horizontally at $a$ m s$^{-2}$ under the action of these forces.



(a) Show that

$$P = 125\left(\frac{a}{\cos 80^\circ} + \frac{g}{\sin 80^\circ}\right)$$

**(5)**

(b) Find the value of $a$ for which $Q$ is zero.

**(3)**
**(Total 8 marks)**

## Q10.

A boat moves with constant acceleration, so that its velocity $\mathbf{v}$ m s$^{-1}$ at time $t$ seconds is given by

$$\mathbf{v} = (4.5 - 0.9t)\mathbf{i} + (2 + 0.5t)\mathbf{j}$$

where the unit vectors **i** and **j** are directed east and north respectively.

(a) Find the velocity of the boat when $t = 0$.

**(1)**

(b) Find the velocity of the boat when $t = 5$.

**(1)**

(c) State the direction in which the boat is travelling when $t = 5$.

**(1)**

(d) Find the acceleration of the boat.

**(3)**

(e) The mass of the boat is 250 kg. Find the magnitude of the resultant force acting on the boat.

**(3)**

(f) Draw a diagram to show the direction of the resultant force on the boat and calculate the angle between this force and the unit vector **i**.

**(4)**

**(Total 13 marks)**

## Q11.

Two forces, **P** = (6**i** − 3**j**) newtons and **Q** = (3**i** + 15**j**) newtons, act on a particle. The unit vectors **i** and **j** are perpendicular.

(a) Find the resultant of **P** and **Q**.

**(2)**

(b) Calculate the magnitude of the resultant of **P** and **Q**.

**(2)**

(c) When these two forces act on the particle, it has an acceleration of (1.5**i** + 2**j**) m $s^{-2}$. Find the mass of the particle.

**(2)**



(d) The particle was initially at rest at the origin.

(i) Find an expression for the position vector of the particle when the forces have been applied to the particle for $t$ seconds.

**(2)**

(ii) Find the distance of the particle from the origin when the forces have been applied to the particle for 2 seconds.

**(2)**

**(Total 10 marks)**

## Q12.

A block, of mass 5 kg, slides down a rough plane inclined at 40° to the horizontal. When modelling the motion of the block, assume that there is no air resistance acting on it.

(a) Draw and label a diagram to show the forces acting on the block.

**(1)**

(b) Show that the magnitude of the normal reaction force acting on the block is 37.5 N, correct to three significant figures.

**(2)**

(c) Given that the acceleration of the block is 0.8 m $s^{-2}$, find the coefficient of friction between the block and the plane.

**(6)**

(d) In reality, air resistance does act on the block. State how this would change your value for the coefficient of friction and explain why.

**(2)**

**(Total 11 marks)**

## Q13.

A trolley, of mass 100 kg, rolls at a constant speed along a straight line down a slope inclined at an angle of 4° to the horizontal.

Assume that a constant resistance force, of magnitude $P$ newtons, acts on the trolley as it moves. Model the trolley as a particle.

(a) Draw a diagram to show the forces acting on the trolley.

**(1)**

(b) Show that $P$ = 68.4 N, correct to three significant figures.

**(3)**

(c) (i) Find the acceleration of the trolley if it rolls down a slope inclined at 5° to the horizontal and experiences the same constant force of magnitude $P$ that you found in part (b).

**(4)**

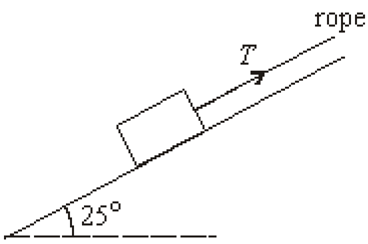
(ii) Make one criticism of the assumption that the resistance force on the trolley is constant.

**(1)**

**(Total 9 marks)**

## Q14.

A rough slope is inclined at an angle of 25° to the horizontal. A box of weight 80 newtons is on the slope. A rope is attached to the box and is parallel to the slope. The tension in the rope is of magnitude $T$ newtons. The diagram shows the slope, the box and the rope.




(a) The box is held in equilibrium by the rope.

(i) Show that the normal reaction force between the box and the slope is 72.5 newtons, correct to three significant figures.

**(3)**

(ii) The coefficient of friction between the box and the slope is 0.32. Find the magnitude of the maximum value of the frictional force which can act on the box.

**(2)**

(iii) Find the least possible tension in the rope to prevent the box from moving down the slope.

**(4)**

(iv) Find the greatest possible tension in the rope.

**(3)**

(v) Show that the mass of the box is approximately 8.16 kg.

**(1)**

(b) The rope is now released and the box slides down the slope. Find the acceleration of the box.

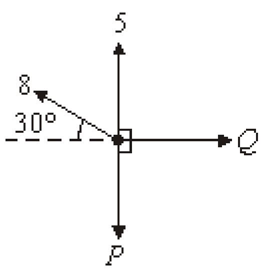
**(3)**

**(Total 16 marks)**



## Q15.

A particle is in equilibrium under the action of four horizontal forces of magnitudes 5 newtons, 8 newtons, $P$ newtons and $Q$ newtons, as shown in the diagram.




(a) Show that $P = 9$.

**(3)**

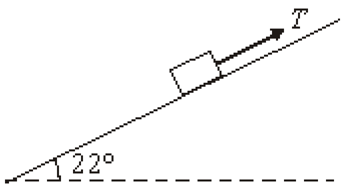
(b) Find the value of $Q$.

**(2)**
**(Total 5 marks)**

## Q16.

A block is being pulled up a rough plane inclined at an angle of 22° to the horizontal by a rope parallel to the plane, as shown in the diagram.

The mass of the block is 0.7 kg, and the tension in the rope is $T$ newtons.




(a) Draw a diagram to show the forces acting on the block.

**(1)**

(b) Show that the normal reaction force between the block and the plane has magnitude 6.36 newtons, correct to three significant figures.

**(3)**

(c) The coefficient of friction between the block and the plane is 0.25. Find the magnitude of the frictional force acting on the block during its motion.

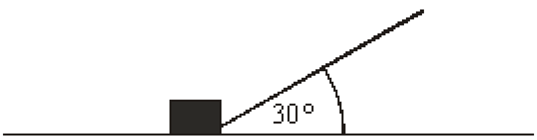
**(2)**

(d) The tension in the rope is 5.6 newtons. Find the acceleration of the block.

**(4)**
**(Total 10 marks)**

## Q17.

The diagram shows a rope that is attached to a box of mass 25 kg, which is being pulled along rough horizontal ground. The rope is at an angle of 30° to the ground. The tension in the rope is 40 N. The box accelerates at 0.1 m $s^{-2}$.




(a) Draw a diagram to show all of the forces acting on the box.

**(1)**

(b) Show that the magnitude of the friction force acting on the box is 32.1 N, correct to three significant figures.

**(3)**

(c) Show that the magnitude of the normal reaction force that the ground exerts on the box is 225 N.

**(3)**

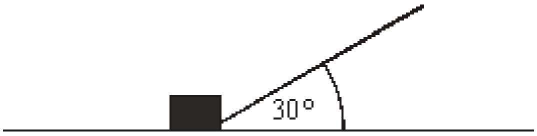
(d) Find the coefficient of friction between the box and the ground.

**(2)**

**(Total 9 marks)**

## Q18.

The diagram shows a rope that is attached to a box of mass 25 kg, which is being pulled along rough horizontal ground. The rope is at an angle of 30° to the ground. The tension in the rope is 40 N. The box accelerates at 0.1 m $s^{-2}$.




(a) Draw a diagram to show all of the forces acting on the box.

**(1)**

(b) Show that the magnitude of the friction force acting on the box is 32.1 N, correct to three significant figures.

**(3)**

(c) Show that the magnitude of the normal reaction force that the ground exerts on the box is 225 N.

**(3)**

(d) Find the coefficient of friction between the box and the ground.

**(2)**

(e) State what would happen to the magnitude of the friction force if the angle between the rope and the horizontal were increased. Give a reason for your answer.

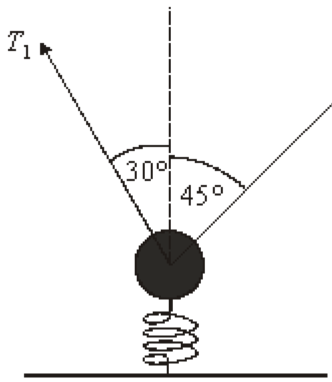
**(2)**

**(Total 11 marks)**



## Q19.

Two ropes are attached to a load of mass 500 kg. The ropes make angles of 30° and 45° to the vertical, as shown in the diagram. The tensions in these ropes are $T_1$ and $T_2$ newtons. The load is also supported by a vertical spring.



The system is in equilibrium and $T_1$ = 200.

(a) Show that $T_2$ = 141, correct to three significant figures.

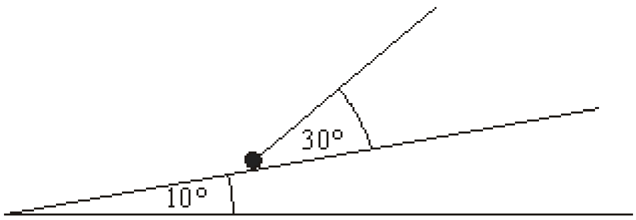
**(3)**

(b) Find the force that the spring exerts on the load.

**(4)**
**(Total 7 marks)**

## Q20.

A rough slope is inclined at an angle of 10° to the horizontal. A particle of mass 6 kg is on the slope. A string is attached to the particle and is at an angle of 30° to the slope. The tension in the string is 20 N. The diagram shows the slope, the particle and the string.




The particle moves up the slope with an acceleration of 0.4 m $s^{-2}$.

(a) Draw a diagram to show the forces acting on the particle.

**(1)**

(b) Show that the magnitude of the normal reaction force is 47.9 N, correct to three significant figures.

**(4)**

(c) Find the coefficient of friction between the particle and the slope.

**(6)**
**(Total 11 marks)**



## Q21.

A car moves in a straight line up a hill which is inclined at an angle of α to the horizontal, where sin α = $\frac{1}{14}$ . The mass of the car is 1200 kg.

(a) During the first part of its motion, the car moves with constant speed and is subject to a propulsive force of 1500 newtons and a resistance force of $R$ newtons.

(i) Draw a diagram showing the forces acting on the car.

**(2)**

(ii) Show that $R$ = 660.

**(4)**

(iii) The car travels 60 metres in 4 seconds. Write down the speed of the car.

**(1)**

(b) During the second part of its motion, the propulsive force ceases to act and the resistance force is still 660 newtons. The car continues to move up the hill until it comes to rest.

(i) Find the magnitude of the retardation of the car in this stage.

**(2)**

(ii) Find the time between the propulsive force ceasing to act and the car coming to rest.

**(2)**
**(Total 11 marks)**

## Q22.

A mountain railway train moves on a straight track. The mass of the train and its passengers is 1000 kg. During its motion the train moves under the action of a variable propulsive force, $P$ newtons, and a constant resistance force of $R$ newtons.

(a) During the first stage of its motion, the train moves horizontally with acceleration 0.25 m s$^{-2}$. In this stage, the value of $P$ is 1200.

Show that $R$ = 950.

**(3)**

(b) During the second stage of its motion, the train moves up a slope inclined at an angle $\alpha$ to the horizontal, where sin $\alpha$ = 0.1.




In this stage, the value of $P$ = 2100, and the value of $R$ remains at 950. Find the acceleration of the train.

**(4)**
**(Total 7 marks)**

## Q23.

Two particles, $A$ and $B$, are connected by a light inextensible string which passes over a smooth, fixed peg, as shown in the diagram.


The particle $A$, of mass 0.6 kg, is in contact with a smooth horizontal surface, and the particle $B$, of mass 0.1 kg, hangs freely above the ground. The system is released from rest with the string taut and $A$ moves towards the peg.

It can be assumed that, during the subsequent motion, $A$ does **not** reach the peg.

(a) While the particles move freely, the string is taut.

(i) Show that the acceleration of the particles is 1.4 m $s^{-2}$.

**(5)**

(ii) Find the tension in the string.

**(1)**

(iii) Find the magnitude of the resultant force on the peg due to the tension in the string.

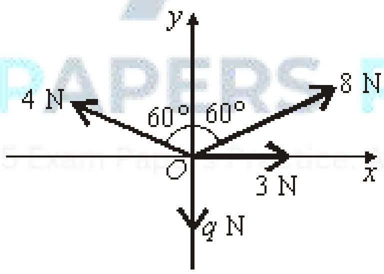
**(3)**

(b) After $q$ seconds of the motion, the particle $B$ hits the ground and remains there. The string connecting $A$ and $B$ slackens and $A$ continues to move towards the peg. Sketch a velocity–time graph to show the two stages of the motion of $A$ after being released from rest.

**(3)**
**(Total 12 marks)**

## Q24.

A particle is at a point $O$ on a smooth horizontal surface. It is acted on by four horizontal forces of magnitudes 3 N, 8 N, 4 N and $q$ N. The directions of these forces, relative to the horizontal axes $Ox$ and $Oy$, are shown in the diagram below.




The resultant, **R** newtons, of these forces acts along the line $Ox$.

(a) Show that $q$ = 6.

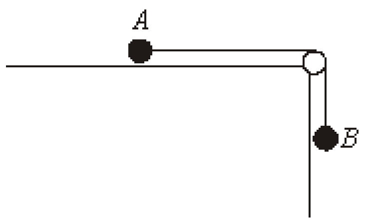
**(4)**

(b) Find the magnitude of **R**.

**(3)**
**(Total 7 marks)**

## Q25.

Two particles, $A$ and $B$, are connected by a light, inextensible string which passes over a smooth, fixed peg, as shown in the diagram.




The particle $A$ is of mass 0.5 kg and the particle $B$ is of mass 0.1 kg. The particle $A$ is in contact with a rough horizontal surface.

(a) The system is at rest and the particle $B$ hangs vertically.

(i) Show that the tension in the string is 0.98 newtons.

**(1)**

(ii) The particle $A$ rests in limiting equilibrium. Show that the coefficient of friction between $A$ and the surface is 0.2.

**(3)**

(iii) Draw a diagram to show the forces acting on the peg due to the string.

**(1)**

(iv) Find the magnitude of the resultant force on the peg due to the string.

**(2)**

(b) An additional mass of 0.1 kg is attached to the particle $B$ and the system is released from rest. The particle $A$ moves across the surface towards the peg and $B$ moves vertically downwards. The particles move with acceleration $a$ m s$^{-2}$ and the tension in the string is $T$ newtons.



(i) Show that $a$ = 1.4.



**(5)**

(ii) Find the value of $T$.

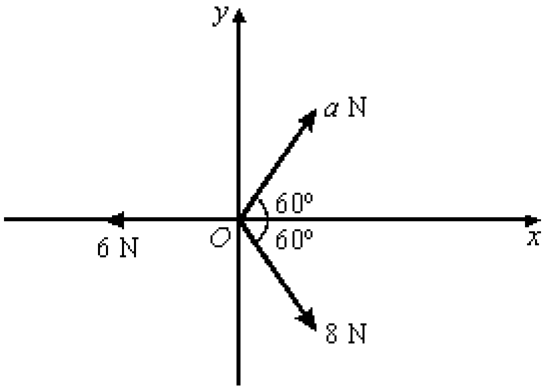
**(1)**

(iii) After falling a distance of 0.7 metres, $B$ hits the ground. Find the time between the particles being released and $B$ hitting the ground.

**(2)**

**(Total 15 marks)**

**Q26.**




A particle is at a point $O$ on a smooth horizontal surface. It is acted on by three horizontal forces of magnitudes 6 N, 8 N and  N. Relative to horizontal axes $Ox$ and $Oy$, the directions of these three forces are shown in the diagram. The resultant, **R**, of these forces acts along the line $Oy$.

(a) Show that $a = 4$.

**(4)**

(b) Find the magnitude of **R**.

**(3)**
**(Total 7 marks)**

## Q27.

A child travels down a slide at a **constant speed**. Model the slide as a rough plane inclined at an angle of 40° to the horizontal. Model the child as a particle of mass 20 kg and assume that there is no air resistance as the child moves.

(a) (i) Draw a diagram to show the forces acting on the child.

**(1)**



(ii) Show that the magnitude of the normal reaction force acting on the child is approximately 150 N.



**(2)**

(iii) Find the magnitude of the friction force acting on the child and show that the coefficient of friction between the child and the slide is 0.84, correct to two significant figures.

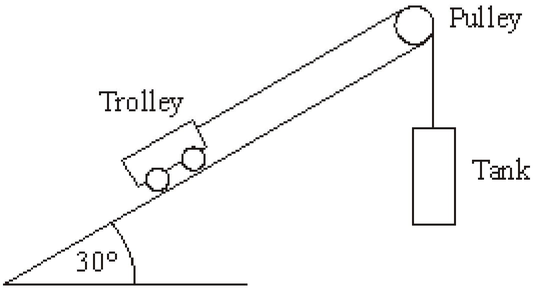
**(4)**

(b) In reality, air resistance acts on the child as she slides at a constant speed. How would this affect the value of the coefficient of friction that you calculated in part (a)(iii)?

**(1)**
**(Total 8 marks)**

## Q28.

A trolley, of mass 200 kg, is on a slope inclined at 30° to the horizontal. It is attached to a tank by a light, inextensible rope that passes over a smooth, light pulley. The tank can be filled with water. The situation is shown in the diagram below.




Model the tank and trolley as particles and assume that there is no resistance to motion.

(a) When the tank is filled with water, the total mass of the tank and the water is 150 kg.

Show that the acceleration of the tank and trolley is 1.4 m s$^{-2}$ .

**(7)**

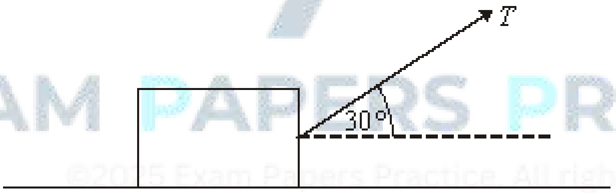
(b) When the tank is empty, the trolley can be set in motion, so that it moves with a constant speed up the slope. Find the mass of the empty tank.

**(4)**

**(Total 11 marks)**

## Q29.

A heavy crate, of mass 200 kg, is pulled along a rough horizontal surface at a constant speed by a rope. The rope is at an angle of 30° to the horizontal. The tension in the rope is $T$ newtons. The coefficient of friction between the crate and the surface is 0.6. Model the crate as a particle.




(a) Draw a diagram to show the forces acting on the crate.

**(1)**

(b) Show that the magnitude of the normal reaction force on the crate is (1960 – 0.5 $T$) newtons.

**(3)**

(c) Find $T$.

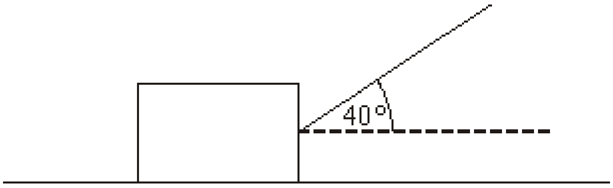
**(5)**

**(Total 9 marks)**

## Q30.

A box, of mass 50 kg, is dragged along rough horizontal ground by a rope attached to the box. The rope makes an angle of 40° with the ground and the tension in the rope is $T$

newtons. The coefficient of friction between the box and the ground is 0.6. The diagram shows the box and the rope.




Model the box as a particle.

(a) Draw a diagram to show the forces acting on the box.

**(1)**

(b) Find, in terms of $T$, the magnitude of the normal reaction force acting on the box.

**(3)**

(c) Show that the magnitude of the friction force acting on the box is

$$294 - 0.6T \sin 40°$$

**(2)**

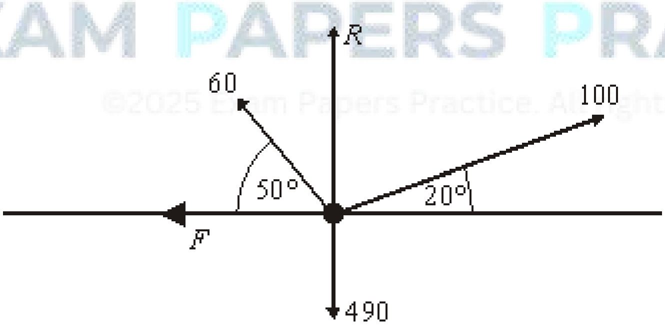
(d) The box accelerates at 0.5 m s$^{-2}$. Find $T$.

**(4)**

**(Total 10 marks)**

## Q31.

A box, of mass 50 kg, moves in a straight line on a rough horizontal surface. The diagram below shows **all** the forces acting on the box, as it moves. The magnitude of each force is in newtons. The box is modelled as a particle.




Note that the weight of the box has been included in the diagram.

(a) Find $R$, the magnitude of the normal reaction force acting on the box.

**(3)**

(b) The box accelerates at 0.5 m s$^{-2}$.

(i) Show that $F$, the magnitude of the friction force acting on the box, is approximately 30.4.

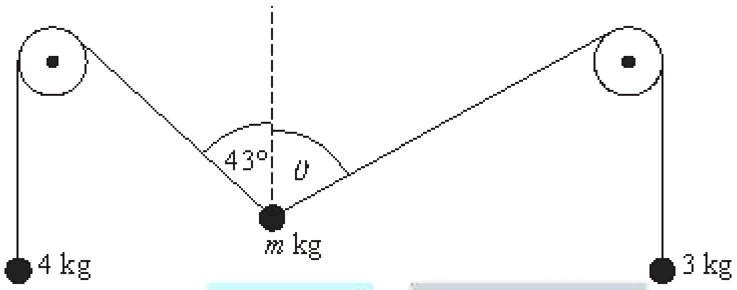
**(4)**

(ii) Find the coefficient of friction between the box and the surface.

**(2)**
**(Total 9 marks)**

## Q32.

Two light, inextensible strings are attached to a particle of mass $m$ kg. Each string passes over a fixed, smooth, light pulley. The other end of one string is attached to a particle of mass 4 kg. The other end of the second string is attached to a particle of mass 3 kg. The diagram shows the system in its equilibrium position. The angles marked on the diagram are between the strings and the vertical.




(a) Calculate the tension in each of the strings.

**(2)**

(b) Show that $\theta$ = 65.4°, correct to three significant figures.

**(5)**

(c) Find $m$.

**(4)**
**(Total 11 marks)**



## Q33.

A rope is used to pull a box, of mass 20 kg, up a rough slope. Assume that the box moves with a constant acceleration.

(a) As the box moves 4 metres up the slope its speed increases from 1 m $s^{-1}$ to 5 m $s^{-1}$. Calculate the acceleration of the box.

**(3)**

(b) The slope is at an angle of 30° to the horizontal. Assume that the rope is parallel to the slope. The coefficient of friction between the box and the slope is 0.4.

(i) Draw a diagram to show the forces on the box.

**(1)**

(ii) Show that the magnitude of the friction force acting on the box is approximately 68 N.

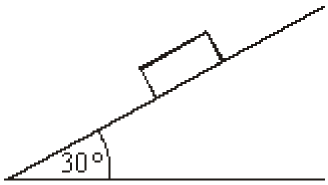
**(3)**

(iii) Find the tension in the rope.

**(4)**
**(Total 11 marks)**

## Q34.

A block, of mass 7 kg, is placed on a rough slope that is inclined at 30° to the horizontal, as shown in the diagram. The block remains at rest in this position.




(a) Draw a diagram to show the forces acting on the block.

**(1)**

(b) Find the magnitude of the normal reaction force acting on the block.

**(2)**

(c) Find the magnitude of the friction force acting on the block.

**(2)**

(d) The coefficient of friction between the block and the plane is $\mu$. Find an inequality that $\mu$ must satisfy.

**(2)**

(e) A similar block, of mass 14 kg, is placed on the slope. Does this block remain at rest or slide? Give a reason for your answer.

**(2)**
**(Total 9 marks)**